

MIKE HOLZINGER

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Brooklyn, NY 11213

Skills and Expertise

Languages: Python (7 yrs), C (7 yrs), Ruby (2 yrs), Shell/Bash (10+ yrs), YAML (10 yrs)

Developer Tools & Automation: GitLab CI/CD, GitHub Actions, Jenkins, Ansible, Terraform, Claude Code, MCP (Model Context Protocol)

Containers & Orchestration: Docker, Kubernetes, Helm, ArgoCD

Cloud Platforms: AWS (ECS, S3, SES), GCP (GKE), IBM Cloud

Observability & Monitoring: Grafana, Prometheus, ELK Stack, OpenSearch, OpenTelemetry

Security & Compliance: PKI, FIPS 140-2, Certificate Authority Management, AES Encryption, HashiCorp Vault, LDAP/Active Directory

Databases: MySQL/MariaDB, PostgreSQL, CouchDB, OpenSearch

Operating Systems: Linux/Unix Administration (12+ yrs), Windows Server Administration

EXODUS POINT CAPITAL MANAGEMENT

Sr. Software Engineer (Contract work - Completed)

April 2025 - May 2026

Python Environment Distribution Platform:

- Designed and implemented custom Python distribution solution, reducing deployment time from 30 minutes to under 2 minutes for quantitative analytics teams across the organization
- Built in-house automation platform using GitLab CI/CD, enabling full control and customization of Python environment delivery without vendor dependencies
- Developed automated deployment tooling that integrated with existing Windows infrastructure, providing seamless environment activation and application integration
- Collaborated with Windows administration and quantitative analytics teams to deliver standardized Python environments supporting portfolio management workflows

- Served as primary Python support contact for organization-wide environment standardization

Platform Engineering & Automation:

- Built Kubernetes-based automated processing pipeline using AWS services and infrastructure-as-code
- Implemented LDAP-integrated filtering and indexing system with metadata extraction capabilities
- Developed secure, air-gapped LLM integration for automated content processing
- Managed platform integration and cross-team technical coordination

Key Technologies:

Python, PowerShell, GitLab CI/CD, Docker, Kubernetes, Terraform, ArgoCD, AWS

BAYONE SOLUTIONS

Cisco

Sr. Software Engineer (Contract work - Completed)

April 2024 - October 2024

WebEx Framework Modernization:

- Contributed to modernizing the **WebEx corporate framework**, configuring and deploying custom builds for over **300 clients**, enhancing the platform's flexibility and performance. - **Migrated** the codebase from **Python 2.8 to Python 3.12**, utilizing tools like **Tox** and **Pytest** for testing and validation, ensuring code reliability and compatibility.

Configuration Architecture & Automation:

- Enhanced the configuration architecture by implementing design patterns and **Jinja templating**, improving maintainability and scalability of configuration management. - Developed new **linting tools** to validate **JSON, YAML**, and environment key settings, ensuring consistency across more than 5,000 configuration documents.
- Streamlined the deployment process using **Docker** and **Jenkins**, reducing build times and improving continuous integration workflows.

Collaboration & Testing:

- Collaborated with cross-functional teams to create comprehensive **unit tests** for system functionality, increasing test coverage and software quality.

Key Technologies:

Docker, GitHub, Jenkins, Jinja2, Pytest, Python 3.x, Tox, JSON, YAML

MRI TECHNOLOGIES

NASA

Sr. Software Engineer

March 2023 - November 2023

Cloud Infrastructure Enhancement:

- Managed and refined Kubernetes deployments and GitLab pipelines on Google Cloud Platform, significantly enhancing infrastructure scalability and resilience.
- Transitioned GitLab pipelines from Docker-in-Docker to **Kaniko**, reducing deployment times from **1 hour to under 8 minutes** and eliminating legacy dependencies.

Security & Dependency Management:

- Applied Helm version locking to mitigate security vulnerabilities and improve dependency management, boosting system security and reliability.

Automation & Efficiency:

- Automated CI processes and scripted Kubernetes tasks, including database job scheduling and export operations, contributing to the efficiency and reliability of NASA's cloud platform.

Observability & Team Training:

- Deployed Grafana dashboards and trained development teams on utilizing logging assets and Kubernetes pod metrics, enhancing platform observability and performance monitoring.

Key Technologies:

Docker, GitLab, Google Cloud Platform, Grafana, Helm, Keycloak, Kubernetes, Python, Terraform

HTC GLOBAL

Disney (Parks and Recreation)

Site Reliability Engineer (Contract work - Completed)

September 2022 - April 2023

Application Migration & Management:

- Served as an SRE for over 500 applications migrated from on-premises hardware to AWS ECS-hosted solutions.
- Managed application onboarding, scheduled nightly maintenance windows, and acted as the incident management commander to ensure seamless operations.

Automation & Operational Efficiency:

- Developed Python automation scripts for ServiceNow integration with Slack Notification API,

significantly improving escalation processes and reducing missed change management windows.

- Utilized Python with AppDynamics and Grafana for enhanced monitoring and proactive incident management, contributing to increased application reliability.

Key Technologies:

AWS ECS, AppDynamics, Grafana, OpenTelemetry, Prometheus, Python, Rundeck, ServiceNow, Slack Workflow Builder, Spark

SMARSH

Site Reliability Engineer

July 2021 - September 2022

- **Infrastructure Modernization:** Collaborated with application developers to migrate a .NET-based VMware-hosted application stack to Kubernetes, transitioning legacy Ansible and BOSH infrastructure to a hybrid cloud environment using VMware Cloud Provider Interface and AWS/EKS.
- **Helm Chart Development:** Gained advanced proficiency in Helm for chart development, template management, and deployment strategies, implementing security measures and optimization techniques to enhance operational efficiency and system integrity.
- **Automation & CI/CD:** Streamlined application deployment and management by automating CI/CD pipelines using tools like CircleCI and GitHub Actions. Leveraged Ansible, Docker, and Kubernetes to automate infrastructure tasks and improve scalability.
- **Observability & Monitoring:** Improved system reliability by employing monitoring and observability tools such as OpenTelemetry. Deployed new Grafana dashboards to provide insights into system performance and trained teams on utilizing these metrics effectively.

Key Technologies:

Ansible, Artifactory, BOSH, CircleCI, CredHub, Docker, GitHub Actions, HashiCorp Vault, Helm, Kubernetes, OpenTelemetry, PowerShell, Python, Terraform, SaltStack, vSphere, YAML

IBM

Software Engineer

July 2015 - October 2019

Roles & Responsibilities:

- **Cloudant Watson Data Platform:** Worked on operations and development for Cloudant (CouchDB), identifying and resolving production issues.
- **FoundationDB Document Storage:** Developed metrics performance tooling and contributed to the groundwork of a new document store infrastructure for the Watson Data

Platform.

- **Global Managed Cloud (GBS):** Independent contributor for Oracle SAP and distributed VMware offerings. Created tooling for CI/CD, operations, and enhanced security compliance.
- **IBM Cloud:** Object Storage & OpenStack: DevOps and operations engineer for IBM's Cleversafe acquisition and OpenStack services, managing large-scale cloud deployments and customer instances.

Notable achievements:

- Developed a client tool that reduced operator time by providing insights without server node access.
- Created data health investigation tools aggregating cluster information across nodes.
- Automated host discovery for compliance reporting, improving data remediation processes.
- Led orientation and training for new employees, serving as West Coast Operations Lead.
- Built monitoring dashboards with Grafana and implemented operational tooling for large-scale storage deployments.

Key Technologies:

Ansible, AWS, BASH/Shell, Chef, Cloudant, CouchDB, ELK Stack, Erlang, FoundationDB, GitHub Enterprise, Grafana, IBM Cloud, Kubernetes, Linux, OpenStack, PagerDuty, Python, REST APIs, Ruby, Sensu, SoftLayer, VMware, YAML

BLUE BOX GROUP

Operations Engineer - Linux

July 2014 - July 2015

Cloud Migration & Infrastructure:

- Served as a Linux Operations Engineer and DevOps specialist for OpenVZ and OpenStack private cloud environments.
- **Led the migration** of customer tenant OpenVZ applications, databases, and infrastructure to OpenStack, enhancing scalability and performance.

Database Administration & Support:

- Provided admin and on-call support for MySQL/MariaDB and PostgreSQL instances.
- Built new primary and secondary SQL solutions and implemented disaster recovery strategies to ensure data integrity and availability

Service Ownership & Security:

- Acted as the primary service owner for high-availability load-balanced services, optimizing uptime and reliability.
- Served as the Registrar for the published Certificate Authority (CA) and managed the internal CA, ensuring secure communications and compliance with security standards.
- Maintained the corporate bastion/authentication gateway, enhancing security protocols for internal systems

Monitoring & Performance Optimization:

- Implemented monitoring solutions using Nagios, Zabbix, and Grafana to proactively identify and resolve system issues
- Optimized system performance through effective configuration of HAProxy and Nginx for efficient load balancing and traffic management

Key Technologies:

Apache, Grafana, HAProxy, Nagios, Nginx, OpenVZ, PKI, Perl, Python, Ruby, SSH Key Authentication, YubiKey/Yubico, Zabbix

TOWNSEND SECURITY

Software Engineer

2007 - 2013

Development of Cryptographic Client Libraries:

- Utilized **ANSI C** to develop lightweight client libraries for the **Alliance Key Manager** (Hardware Security Module).
- Created a **TLS client library in C** for partners to interface with the key manager, ensuring secure and efficient communication.
- Focused on clean code practices in config file parsing, TLS connection management using FIPS-compliant OpenSSL modules, and certificate generation.

FIPS 140-2 Certification:

- Contributed to the development of cryptographic libraries that achieved **NIST FIPS 140-2 certification**, ensuring compliance with stringent security standards.
- Implemented AES encryption, PRNG, SHA, and HMAC algorithms in accordance with NIST guidelines.

Cross-Platform Development & Automation:

- Developed cross-platform SaaS client applications compatible with both Windows and Linux environments.
- Utilized **Jenkins** for continuous integration, automating builds triggered by Git commits and creating testing harnesses to validate new builds.

Cross-Platform Development & Automation:

- Built a central **Certificate Authority** for testing, demos, and partner deployments using **Bash** and **OpenSSL**.
- Created metrics testing tools for encryption API IOPS, high availability failover, and scaling on AWS using **Perl** and **Bash**.
- Developed a consumer PKI deployment tool for SSL certificates on Windows/Linux platforms using **OpenSSL** and **Nullsoft**

Key Technologies:

ANSI C, AES, PKI, TLS, FIPS 140-2, OpenSSL, Jenkins, Bash/Shell, Perl, Java, AWS, Git, Linux